

Benchmarking unsupervised anomaly detection in brain MRI: a systematic comparison across training regimes and preprocessing pipelines

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Introduction

Unsupervised anomaly detection (UAD) in brain MRI learns from healthy data and flags lesions as deviations, avoiding pixel-level annotations [1]. Diffusion-based methods report substantial improvements over earlier approaches, yet inconsistent training corpora, contrasts, field strengths, and preprocessing make it difficult to judge whether these improvements reflect architecture or pipeline choices [2]. It remains unclear how much contrast matching, multi-site data pooling, and tissue-segmentation preprocessing matter relative to model design, and whether any current method can reliably detect small lesions. StRegA [3] introduced FSL FAST tissue-segmentation as preprocessing; here we apply it systematically to four non-StRegA architectures alongside ten matched training configurations on healthy data (IXI, MOOD), evaluated on BraTS.

Methods

We compare five UAD methods: StRegA [3], a compact context-encoding VAE with native FSL FAST preprocessing; SCR4AD [4], scale-aware contrastive reverse distillation; MAD-AD [5], a masked diffusion model; cDDPM [6], guided conditional denoising-diffusion; and UCCD [7], unsupervised contrastive analysis on a conditional-diffusion backbone (Figure 1).

Training data. Healthy-only training on IXI (3T/1.5T, T1/T2) and MOOD (T1) in ten matched configurations per method covering single-site, mixed-field-strength, and joint T1+T2 regimes. Evaluation on BraTS (T1, T2).

Preprocessing. Each non-StRegA method was tested with and without StRegA’s FSL FAST tissue-segmented input, using three representative configurations. FAST is native to StRegA.

Metrics. Following [2], we report Slice DICE, Volume DICE, and Slice AUROC stratified by tumour size (small/medium/large) with bootstrapped 95% CIs. As external reference, Figure 2 reproduces five baselines from [2] (PHANES, AutoDDPM, DDPM-S, pDDPM, RA); PHANES was additionally run under FAST preprocessing on three configurations.

Results

Contrast matching was the strongest single factor (Figure 2). T2-trained models on BraTS T2 outperformed T1 counterparts across all methods (mean Slice AUROC T1/T2: StRegA 0.73/0.84; cDDPM 0.81/0.85; UCCD 0.81/0.88). The two conditional-diffusion methods led overall: UCCD reached AUROC 0.883 and Large DICE 0.706 (IXI 1.5T → BraTS T2); cDDPM reached AUROC 0.865 and Large DICE 0.768 (IXI 3T+1.5T → BraTS T2), both well above StRegA, SCR4AD, and MAD-AD under the same conditions.

Multi-site data pooling did not uniformly help. cDDPM improved when combining IXI 3T+1.5T (Large DICE 0.768), but MAD-AD degraded with added heterogeneity (AUROC 0.794 → 0.728 → 0.668 on

T1), and SCRD4AD dropped from 0.721 to 0.590. Conditional-diffusion methods tolerated diversity; the others did not.

Applying StRegA's FAST preprocessing to the four other architectures showed a clear architecture-dependent effect (Figure 3). For SCRD4AD, FAST produced the largest gain in the entire benchmark: Slice DICE rose from 0.348 to 0.754 and Small DICE from near-zero to 0.265 (IXI 3T+1.5T T2), the only configuration in which any method achieved non-trivial small-lesion detection. The diffusion-based methods did not benefit: cDDPM fell from 0.531 to 0.184; MAD-AD stayed at or below baseline; UCCD dropped to 0.000 on its strongest configuration. FAST sharpened SCRD4AD anomaly maps but diffusion methods lost discriminative signal once input was reduced to tissue-class maps. Size stratification (Figures 2, 4) confirmed the large > medium > small ordering, with small-tumour DICE near zero in all raw experiments [2]. FAST+SCRD4AD was the sole exception.

Discussion

Contrast matching between training and evaluation data proved the strongest performance determinant, outweighing architecture in most configurations. Conditional-diffusion methods (cDDPM, UCCD) led consistently under matched conditions. Multi-site pooling improved conditional-diffusion but degraded reverse-distillation and masked-diffusion methods. StRegA's FAST preprocessing, when transferred, exposed a clear interaction with model inductive bias: SCRD4AD benefited dramatically and achieved the only non-trivial small-lesion segmentation in the benchmark, whilst diffusion methods lost signal; FAST is thus informative as a probe of model behaviour, beyond its original role in StRegA. Small-lesion detection remains a near-universal failure in raw-input experiments. Validation is limited to BraTS; multi-site testing is under way.

Conclusion

Across 70+ configurations of five UAD families, upstream choices (contrast matching, training-set composition, preprocessing) shifted model rankings more than architecture. Conditional-diffusion methods gave the strongest raw performance, but the FAST-SCRD4AD interaction yielded the best small-lesion segmentation, an otherwise universal failure. These results argue for reporting UAD performance across a matrix of conditions rather than single-configuration comparisons.

Data and Code Availability

Code is available upon request to S.C. Data are publicly available: IXI (<http://brain-development.org/ixi-dataset/>), MOOD (<http://medicalood.dkfz.de/>), BraTS (<http://braintumorsegmentation.org/>).

References

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Figure Captions

Figure 1. Benchmark workflow: healthy-only training on IXI/MOOD in ten matched configurations per method; two preprocessing pipelines (raw N4 vs StRegA’s FSL FAST, applied to four non-StRegA methods); evaluation on BraTS with size-stratified Slice/Volume DICE and AUROC.

Figure 2. Mega-table. Five UAD methods $\times$ 10 raw + 3 FAST configurations each (StRegA: 10 raw only — FAST is native), plus eight external baselines from Bercea et al. (2025). Slice/Volume DICE, AUROC, and size-stratified metrics. Per-model best in bold blue. IQR = $Q3 - Q1$.

Figure 3. Qualitative comparison on a representative BraTS case: T2 input, FAST segmentation, GT mask, and per-method anomaly maps (raw and FAST) for the five UAD methods and the PHANES external benchmark [2].

Figure 4. Per-volume Volume DICE per method, raw (left) vs FAST (right), including the PHANES external benchmark [2] (MOOD+IXI all). Median marked; StRegA shown as native-FAST.